

Breaking the Limit of Graph Neural Networks by Improving the Assortativity of Graphs with Local Mixing Patterns

KDD '21

CHAE WON PARK

Network Science Lab
Dept. of Artificial Intelligence
The Catholic University of Korea

E-mail: codnjsld@catholic.ac.kr

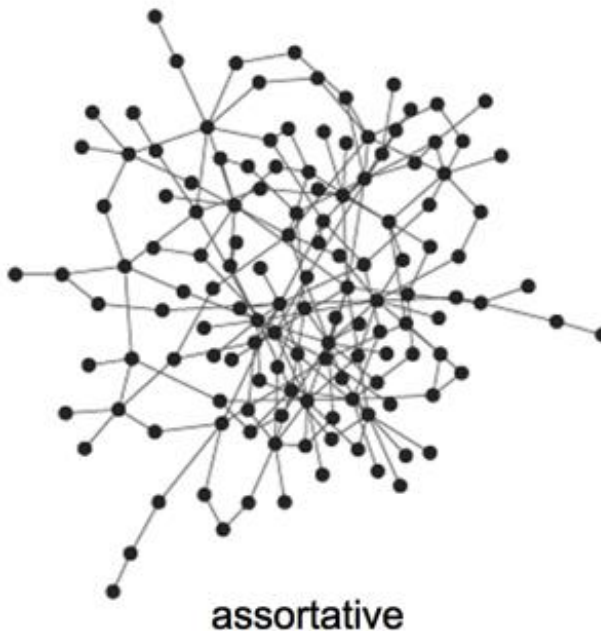
Contents



- Introduction
 - Limitation of GNN
- GNNs and local Mixing
 - Global assortative coefficient
 - Local mixing
- Framework
 - Graph transformation algorithm
 - Computation Graph Transformation
 - Message Passing Mechanism
- Experiments
 - Local Assortativity Distribution Shift
 - Performance of Node Classification
 - Sensitivity Analysis
 - Ablation Analysis

Limitation of GNN

- 최근 GNN 모델들은 message passing을 통해 proximity feature들의 반복적인 aggregation을 기반으로 구축됨
- 예측 성능은 **assortative mixing**에 의해 제한됨
 - Assortative mixing : 비슷한 특징을 가진 노드들끼리 연결되는 성질 (ex. 논문 인용)
 - Disassortative mixing : 서로 다른 특징을 가진 노드들끼리 연결되는 성질 (ex. 포식자/피식자)



Global assortativity coefficient

- Global assortativity coefficient

- 노드들이 유사한 특성이나 label을 가진 다른 노드들과 연결되는 경향

$$r_{\text{global}} = \frac{\sum_g M_{gg} - \sum_g a_g b_g}{1 - \sum_g a_g b_g} \quad (1)$$

$\sum_g \sum_h M_{gh} = 1$ Label g outgoing Label g ingoing

$$\begin{aligned} a_g &= \sum_h M_{gh} \\ b_g &= \sum_h M_{hg} \end{aligned}$$

- 양수 : assortative mixing
- 음수 : disassortative mixing
- 0 : random

$$M_{gh} = \frac{1}{2m} \sum_{i:\tau_i=g} \sum_{j:\tau_j=h} A_{ij} \quad (2)$$

Label g Label h

$$m = |\mathcal{E}|$$

Global assortativity coefficient

- 모든 노드의 mixing pattern이 평균 주변에 집중되어 있을 때만 의미가 있는데, 실제 그래프는 높은 변동성을 나타냄

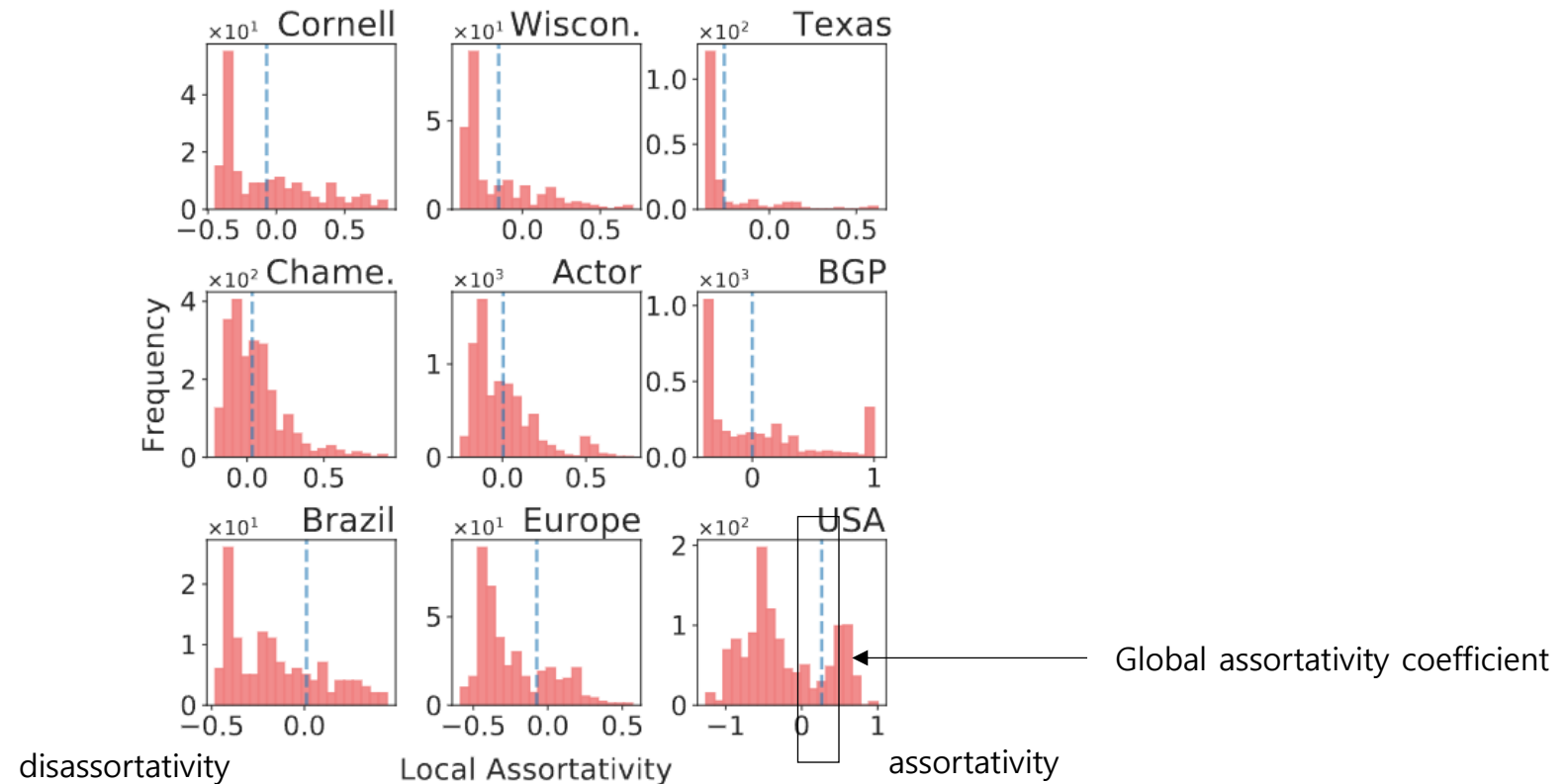


Figure 2: Observed distribution of node level assortativity in various graphs. The blue dotted line indicates global assortativity coefficient.

Local mixing

Node-level assortativity

- 노드 수준에서 측정, random walker를 이용하여 전체 그래프를 방문하는 관점
- local neighborhood의 크기를 변화시켜 개별 노드에서부터 global 수준까지 보간할 수 있게 함

모든 엣지가
동등하게
고려됨 ->
global

$$M_{gh} = \sum_{i:\tau_i=g} \sum_{j:\tau_j=h} \pi_i \frac{A_{ij}}{d_i} \quad (3)$$

Label g -> h

$\pi_i = d_i/2m$ random walk에서
각 노드 i에 머물 확률

$$\pi_i A_{ij} / d_i = 1/2m$$

엣지(i,j)를 통한 이동 확률

Local mixing
matrix

$$M_{gh}(l) = \sum_{i:\tau_i=g} \sum_{j:\tau_j=h} w(i;l) \frac{A_{ij}}{d_i} \quad (4)$$

eq6

Local
assortativity
coefficient

$$r_\alpha(l) = \frac{\sum_g M_{gg}(l) - \sum_g a_g^2}{1 - \sum_g a_g^2} \quad (5)$$

$$w_{tt}(i;l) = \int_0^1 w_\alpha(i;l) d\alpha \quad (6)$$

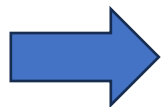
l with a probability of $(1 - \alpha)$

- $a=0$: 시작 노드로 돌아감 (local neighborhood)
- $a=1$: 다시 돌아가지 않고 그래프 전체 탐색 (global assortativity)

모든 α 값에 대해 적분하여 평균냄
-> 다양한 scale에서의 지역적 정보를 균등하게 고려

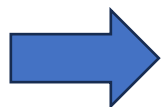
Graph transformation algorithm

- Message passing 기반의 GNN method는 Disassortative mixing에 대해 좋은 representation을 제공하지 못함



더 높은 assortativity를 갖는 새로운 그래프를 생성할 수 있는
graph transformation algorithm을 만들자

- Key idea : 원본 그래프의 구조적, 근접성 정보를 활용
 - 1) 원본 input graph에서 class label 정보를 뺀 노드 집합을 사용한다.
 - 2) 원본 그래프의 model free 방식으로 Structural equivalence를 인코딩한다.
 - 3) 원본 그래프에 보이는 proximity 정보를 인코딩한다.



WRGNN (weighted relational GNN)

Computation Graph

- 노드 간의 구조적 동등성 비교 – 다양한 계층에서 ordered degree sequence 비교
- 두 노드 g, h 사이 **structural distance**

$$f_{\tau}(g, h) = f_{\tau-1}(g, h) + \mathcal{D}(s(\mathcal{N}_{\tau}(g)), s(\mathcal{N}_{\tau}(h))) \quad (7)$$

↑ hop
 ↑ 두 sequence 간의 거리
 ↑ Degree sequence
 ↑ 노드 g 의 τ hop 이웃

- 두 degree sequence가 비슷하면 거리가 가까움
- Hop이 증가함에 따라 거리도 증가함
- 유사도 측정 : FastDTW (Fast Dynamic Time Warping)

Graph Construction

- Weighted multi-relational **computation graph C**

- 노드 쌍 간의 여러 계층에서 Structural distance와 원본 input 그래프 G에 있는 proximity 정보를 인코딩함

Algorithm 1: Construction of computation graph

Input: Original Input Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$;

Hyper-Params.: No. of structural relations $T < \text{dia}(\mathcal{G})$

Output: Computation graph $\mathcal{C}$

```

1 begin
2    $\mathcal{E}' \leftarrow \emptyset$ ;  $n \leftarrow |\mathcal{V}|$ ;
3    $w_\tau \in \mathbb{R}^{n \times n} \leftarrow 0, \forall \tau \in \{0, 1 \dots T\}$ ;  $w_p \in \mathbb{R}^{n \times n} \leftarrow 0$ ;
4    $f_{-1} \leftarrow 0$ ;
5   for  $(g, h) \in \mathcal{V}^2$  do
6     for  $\tau \in \{0, 1 \dots T\}$  do
7        $f_\tau(g, h) \leftarrow f_{\tau-1}(g, h) + \mathcal{D}(s(\mathcal{N}_\tau(g)), s(\mathcal{N}_\tau(h)))$ ;
8        $w_\tau(g, h) \leftarrow e^{-f_\tau(g, h)}$ 
9        $\mathcal{E}' \leftarrow \mathcal{E}' \cup (g, h, \tau)$ ;
10    end
11  end
12  for  $(g, h) \in \mathcal{E}$  do
13     $\mathcal{E}' \leftarrow \mathcal{E}' \cup (g, h, p)$ ;  $w_p(g, h) \leftarrow 1$ 
14  end
15   $\mathcal{R} \leftarrow \{0, 1 \dots T\} \cup p$ ;
16  return  $\mathcal{C} = (\mathcal{V}, \mathcal{E}', \mathcal{R}, \{w_\tau, \forall \tau \in \mathcal{R}\})$ 
17 end
    
```

↑
Hyperparameter (hop)

$$w_\tau(g, h) = e^{-f_\tau(g, h)}, \quad \tau = 0, 1, \dots, T \quad (8)$$

$$(\mathcal{V}, \mathcal{E}', \mathcal{R})$$

$$|\mathcal{R}| = T + 1 + 1$$

- G에서 구조적 거리를 계산한 τ hop 이웃에 해당하는 엣지 타입
- Proximity type edge 1

$$(u, v, \tau) \in \mathcal{E}'$$

- 노드 u에서 v로 가는 edge type τ

Message Passing on the Multi-relational Computation Graph

- Aggregate Function

$$m_u = \sum_{\tau \in \mathcal{R}} \sum_{v \in \mathcal{N}_1^\tau(u)} W_\tau h_v w_\tau(u, v) \alpha_\tau(u, v) \quad (9)$$

eq8 eq11

- Attention coefficients

- 노드 u 가 v 의 정보를 얼마나 중요하게 고려해야 하는지를 나타냄

$$e_{u,v}^\tau = a_\tau(W_\tau h_u, W_\tau h_v) = a^T [W_\tau h_u || W_\tau h_v] \quad (10)$$

- Normalized attention weight

$$\alpha_\tau(u, v) = \frac{\exp(\text{LeakyReLU}(e_{u,v}^\tau))}{\sum_{x \in \mathcal{N}_1^\tau(u)} \exp(\text{LeakyReLU}(e_{u,x}^\tau))} \quad (11)$$

eq10

- Update function

$$h'_u = \sigma \left(W_{\text{self}} h_u + W_{\text{neig}} m_u \right) \quad (12)$$

eq9

Local Assortativity Distribution Shift

- Input graph G 의 disassortative nodes에 초점을 맞춰 변환된 computation graph에서 어떻게 mixing 되는지 관찰
- G 에서 disassortative node들이 **computation graph**에서 **assortativity 증가**

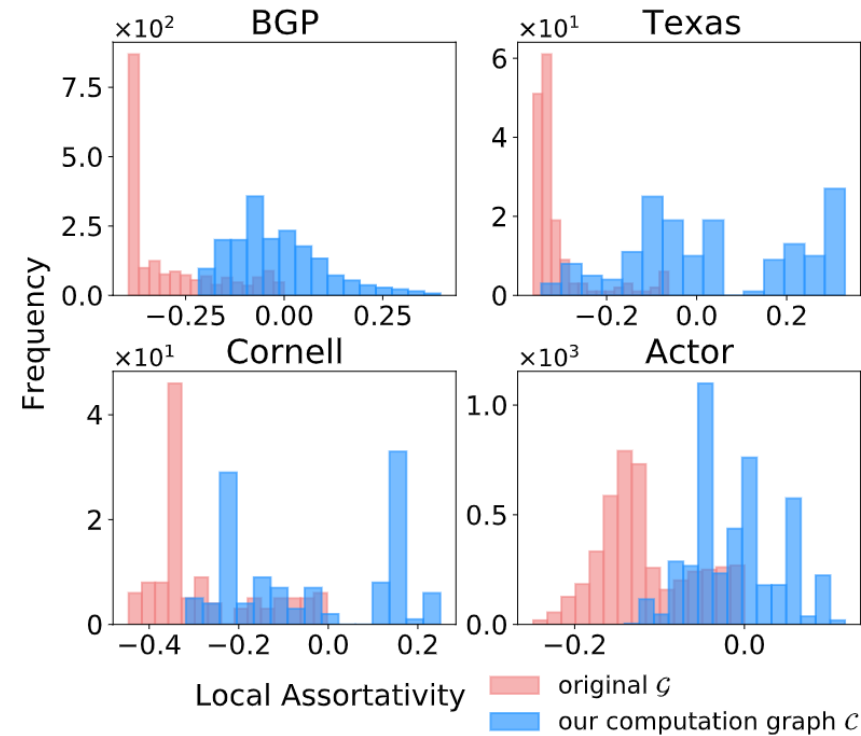
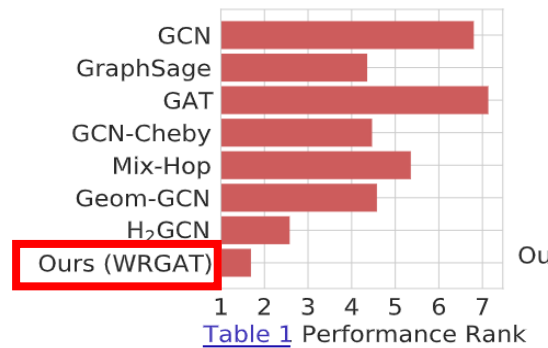


Figure 4: Distributions of node level assortativity for the original graph and the newly constructed computation graph on different datasets.

Performance of Node Classification

Table 1: Semi-supervised node classification showing mean test accuracy $\pm$ std. over 10 runs. Club Suit [♣] denotes result obtained from the best model variant of respective papers.

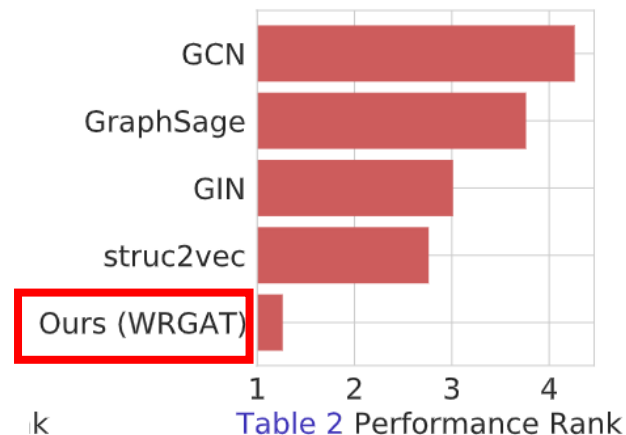
	Chameleon	Squirrel	Actor	Cornell	Texas	Wisconsin	Cora	Citeseer	Pubmed
GCN	59.82 $\pm$ 2.58	36.89 $\pm$ 1.34	30.26 $\pm$ 0.79	57.03 $\pm$ 4.67	59.46 $\pm$ 5.25	59.80 $\pm$ 6.99	87.28 $\pm$ 1.26	76.68 $\pm$ 1.64	87.38 $\pm$ 0.66
GraphSage	58.73 $\pm$ 1.68	41.61 $\pm$ 0.74	34.23 $\pm$ 0.99	75.95 $\pm$ 5.01	82.43 $\pm$ 6.14	81.18 $\pm$ 5.56	86.90 $\pm$ 1.04	76.04 $\pm$ 1.30	88.45 $\pm$ 0.50
GAT	54.69 $\pm$ 1.95	30.62 $\pm$ 2.11	26.28 $\pm$ 1.73	58.92 $\pm$ 3.32	58.38 $\pm$ 4.45	55.29 $\pm$ 8.71	86.37 $\pm$ 1.69	75.46 $\pm$ 1.72	87.62 $\pm$ 0.42
GCN-Cheby	55.24 $\pm$ 2.76	43.86 $\pm$ 1.64	34.11 $\pm$ 1.09	74.32 $\pm$ 7.46	77.30 $\pm$ 4.07	79.41 $\pm$ 4.46	86.86 $\pm$ 0.96	76.25 $\pm$ 1.76	88.08 $\pm$ 0.52
MixHop	60.50 $\pm$ 2.53	43.80 $\pm$ 1.48	32.22 $\pm$ 2.34	73.51 $\pm$ 6.34	77.84 $\pm$ 7.73	75.88 $\pm$ 4.90	83.10 $\pm$ 2.03	70.75 $\pm$ 2.95	80.75 $\pm$ 2.29
Geom-GCN ♣	60.90	38.14	31.63	60.81	67.57	64.12	85.27	77.99	90.05
H ₂ GCN ♣	59.39 $\pm$ 1.98	37.90 $\pm$ 2.02	35.86 $\pm$ 1.03	82.16 $\pm$ 4.80	84.86 $\pm$ 6.77	86.67 $\pm$ 4.69	87.67 $\pm$ 1.42	76.72 $\pm$ 1.50	88.50 $\pm$ 0.64
Ours (WRGAT)	65.24 $\pm$ 0.87	48.85 $\pm$ 0.78	36.53 $\pm$ 0.77	81.62 $\pm$ 3.90	83.62 $\pm$ 5.50	86.98 $\pm$ 3.78	88.20 $\pm$ 2.26	76.81 $\pm$ 1.89	88.52 $\pm$ 0.92



Performance of Node Classification

Table 2: Node classification on Air Traffic Networks and BGP Network. Mean test accuracy $\pm$ std. is shown over 20 runs.

	Brazil	Europe	USA	BGP
GCN	64.55 $\pm$ 4.18	54.83 $\pm$ 2.69	56.58 $\pm$ 1.11	53.33 $\pm$ 0.18
GraphSage	70.65 $\pm$ 5.33	56.29 $\pm$ 3.21	50.85 $\pm$ 2.83	65.19 $\pm$ 0.28
GIN	71.89 $\pm$ 3.60	57.05 $\pm$ 4.08	58.87 $\pm$ 2.12	49.51 $\pm$ 1.52
Struc2vec	70.88 $\pm$ 4.26	57.94 $\pm$ 4.01	61.92 $\pm$ 2.61	48.40 $\pm$ 1.39
Ours	76.92 $\pm$ 5.45	57.12 $\pm$ 2.81	63.02 $\pm$ 1.87	66.54 $\pm$ 0.48



[Table 2](#) Performance Rank

Sensitivity Analysis

Analysis for disassortative networks

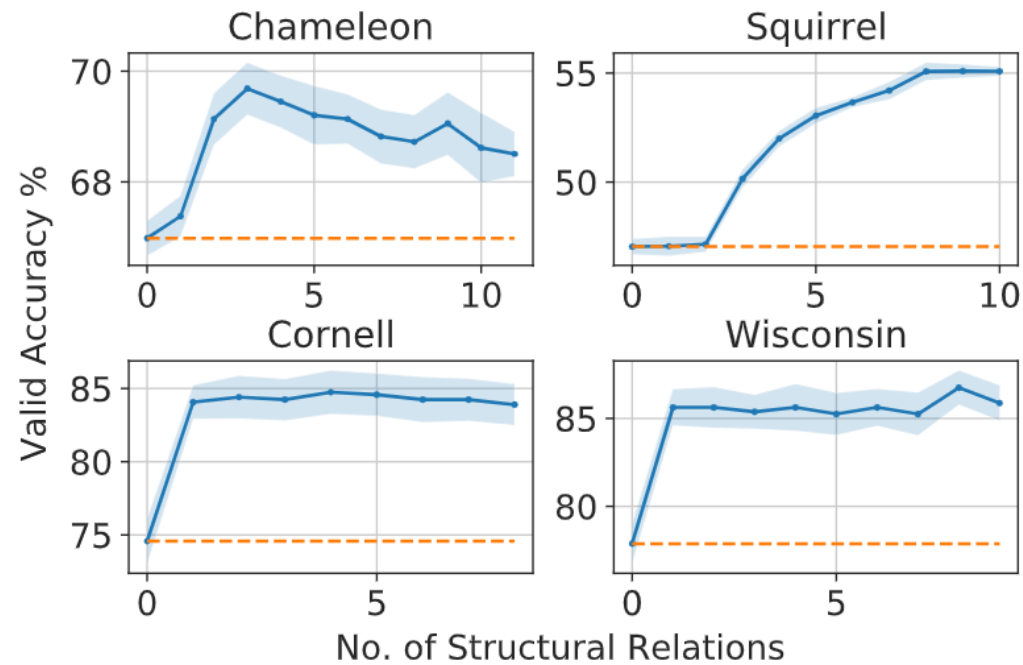


Figure 6: Sensitivity w.r.t structure relations. Baseline is using only proximity information.

Analysis for assortative networks

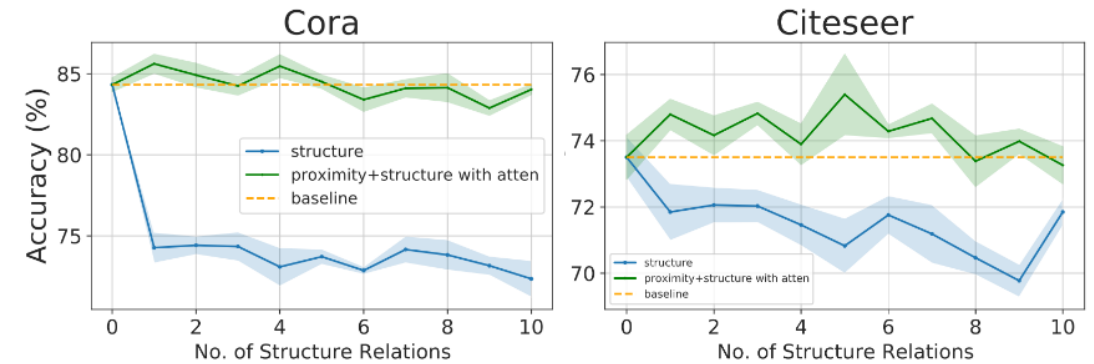


Figure 7: Sensitivity w.r.t structure relations. Baseline is using only proximity information.

Ablation Analysis

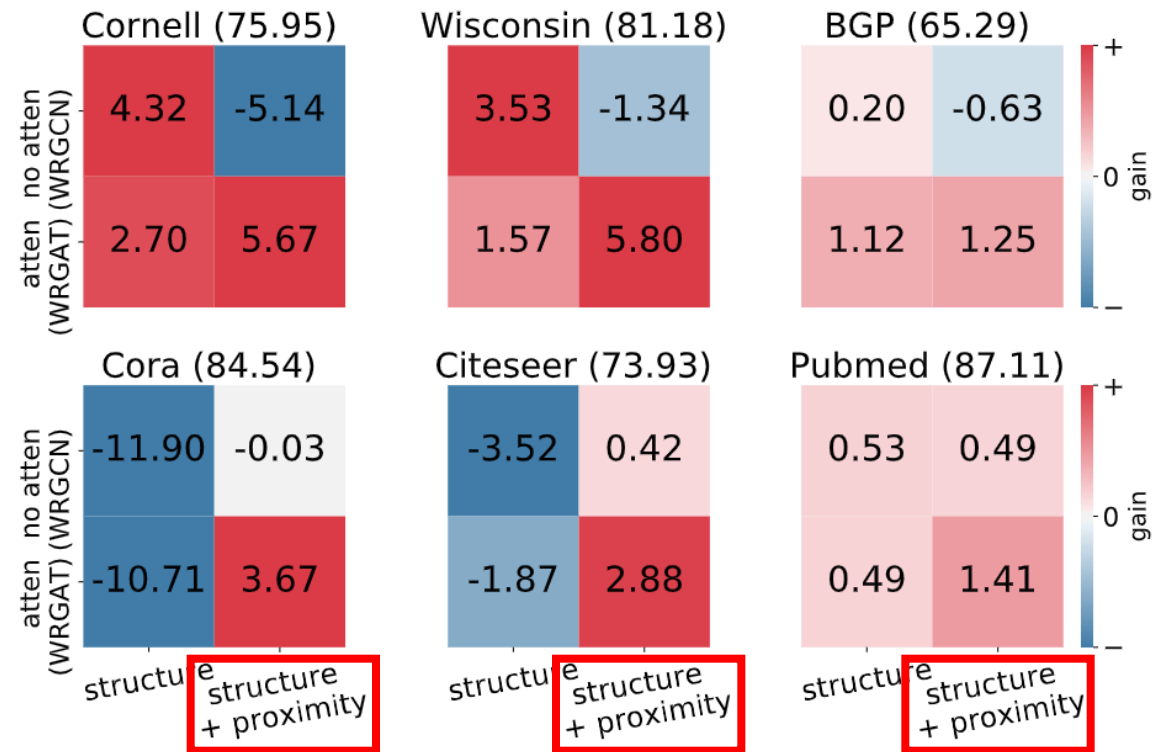


Figure 8: Ablation w.r.t different model and graph information choices. Baseline (proximity only) is shown in parenthesis. Numbers indicate gain over baseline.

Ablation Analysis

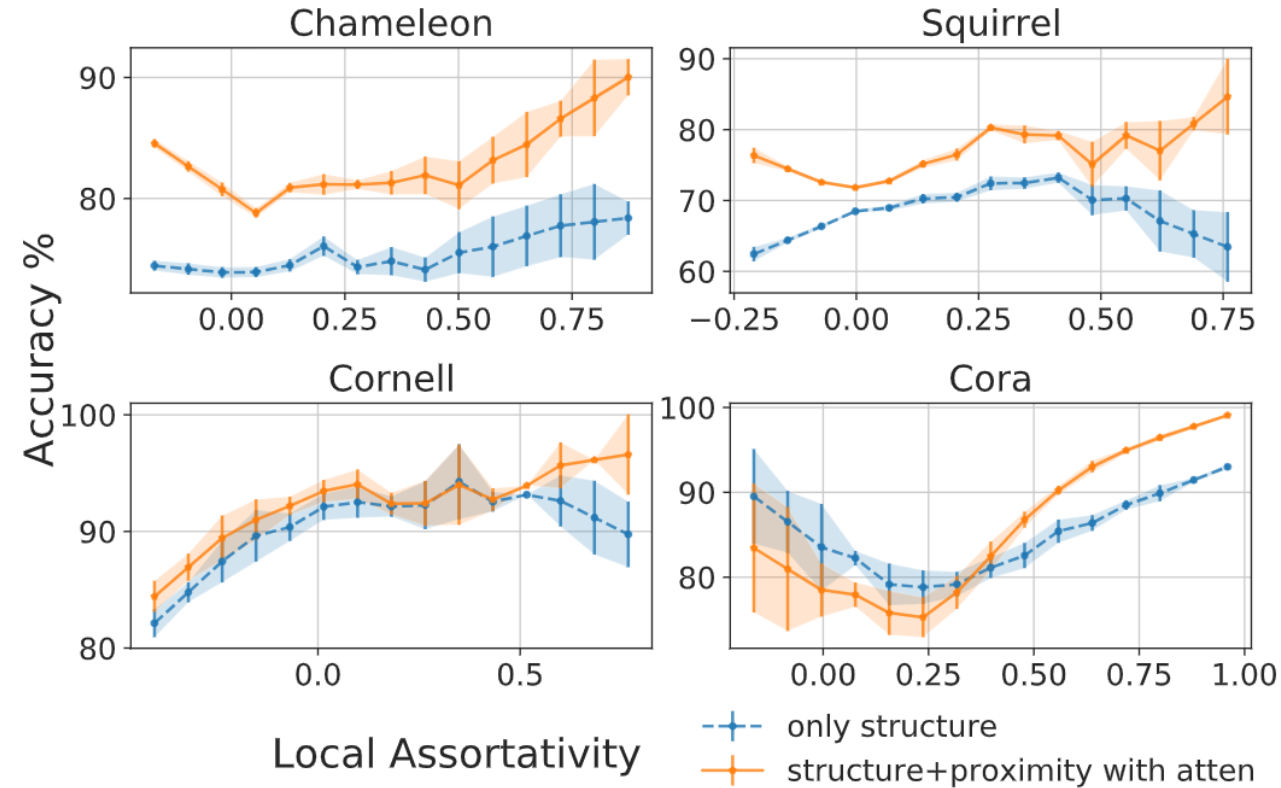


Figure 9: Ablation w.r.t graph info and local assortativity

Local Mixing Patterns

- GNN의 한계에서 벗어나 **local mixing patterns** 관점을 제시함.
- Structure equivariance를 활용하여 associativity를 높이는 **graph transformation algorithm**을 개발함
- GNN을 변환된 computation graph에서 **structure equivariance** 정보와 **proximity** 정보를 적응적으로 선택하여 실행하는 접근 방식이 다양한 실제 네트워크에서 효과적임을 증명함

